

# SYED MUTAHIR HUSSAIN

## AI & Software Engineer

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### PROFESSIONAL SUMMARY

Software & AI Engineer specializing in the intersection of Applied Artificial Intelligence and Core Systems Architecture. Proven track record in building scalable, algorithmically efficient solutions ranging from low-latency edge firmware to enterprise-grade LLM orchestration. Combines rigorous mathematical foundations with practical systems engineering to deploy production-ready intelligence.

### PROFESSIONAL EXPERIENCE

#### Freelance Data Science & AI Consultant | Independent | 2023 – Present

- Architect and deploy end-to-end Machine Learning pipelines and Generative AI solutions for diverse international clients.
- Develop automated Natural Language Processing workflows for large-scale data extraction and sentiment analysis.
- Provide technical consulting on model selection, hyperparameter tuning, and deployment architectures.

#### Cinematic Animator Intern | Unity Development | 2024

- Engineered custom animation states and programmatic scene transitions within the Unity Engine.
- Optimized rendering pipelines, maximizing framerates and ensuring stable performance across diverse hardware constraints.

### SELECTED ENGINEERING PROJECTS

#### Archimedes RAG Engine (LLM / Vector Math)

- Engineered an enterprise-grade Retrieval-Augmented Generation (RAG) pipeline utilizing Pinecone and LangChain.
- Implemented hybrid search alongside custom re-ranking heuristics, increasing context retrieval precision by 40% on massive proprietary datasets.

#### Vigilant Prime Smart Helmet (IoT / Embedded AI)

- Architected a safety-critical IoT motorcycle helmet system focused on autonomous accident detection.
- Deployed quantized convolutional neural networks onto ESP32 hardware via TinyML, achieving sub-45ms offline inference latency through optimized interrupt handling.

#### Titan Vision Analytics (Computer Vision / CUDA)

- Built a distributed Computer Vision pipeline for high-speed object tracking and industrial safety compliance.
- Optimized a custom-trained YOLOv10 model for low-light conditions, executing real-time 4k/60fps streams utilizing hardware acceleration.

#### Spectre MLOps Monitor (DevOps / Infra)

- Designed an automated model observability suite to monitor production AI for feature and concept drift.
- Configured automated retraining triggers utilizing Prometheus and FastAPI when model accuracy thresholds dropped below defined operational parameters.

#### Aura Dashboard (Systems Architecture)

- Engineered a locally-hosted desktop system monitoring application in Python.
- Implemented real-time OS telemetry visualization and a custom high-contrast UI architecture.

### TECHNICAL COMPETENCIES

- Artificial Intelligence:** Deep Learning (PyTorch), NLP & LLMs (LangChain), Computer Vision (YOLOv10, OpenCV), Statistical ML (XGBoost, Scikit-Learn), Vector Databases (Pinecone, Milvus).
- Systems & Software:** C/C++, Python, TypeScript, Object-Oriented Design (SOLID/DRY), Algorithm Design (Space-Time Optimization), Relational/NoSQL Databases (B-Tree+ indexing).
- Infrastructure & Ops:** Real-time Model Observability, Edge AI Deployment (TinyML), Containerization (Docker), REST APIs (FastAPI).

### EDUCATION

#### University of Engineering and Technology (UET), Taxila Bachelor of Science in Computer Science | Expected Graduation: 2026

- Current Status:** 8th Semester
- Core Coursework:** Advanced Data Structures & Algorithms, Object-Oriented Programming, Theory of Automata, Computer Architecture, Operating Systems, Digital Logic Design, Artificial Intelligence, Information Security.